AI & Google Cloud Platform

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Le Monde, Casino, Channel, Club Med, Pernod Ricard, Suez

Dauphine Resources: https://bastinflorian.github.io/teaching/

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Module Overview:

I. Introduction to GCP
II. Databases
III. Application deployment
IV. Vertex AI
V. Security
VI. Application monitoring
VII. GCP Generative AI

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Cloud Skill boost courses

- Deploy a Streamlit App Integrated with Gemini Pro on Cloud Run
- Build an LLM and RAG-based Chat Application with AlloyDB and Vertex AI
- Enhance Text Generation with RAG, LangChain, and Vertex AI

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I. Introduction to GCP

Google Cloud

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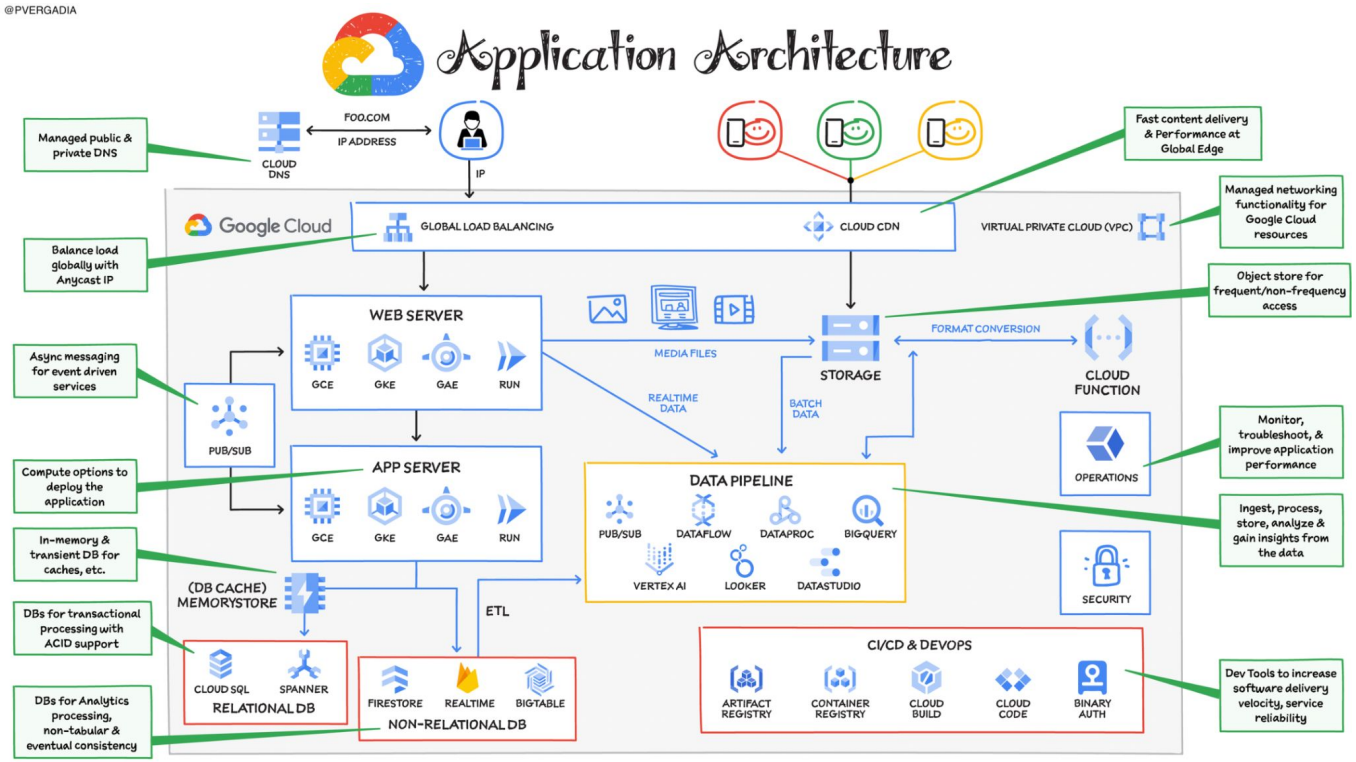
I. Introduction to GCP



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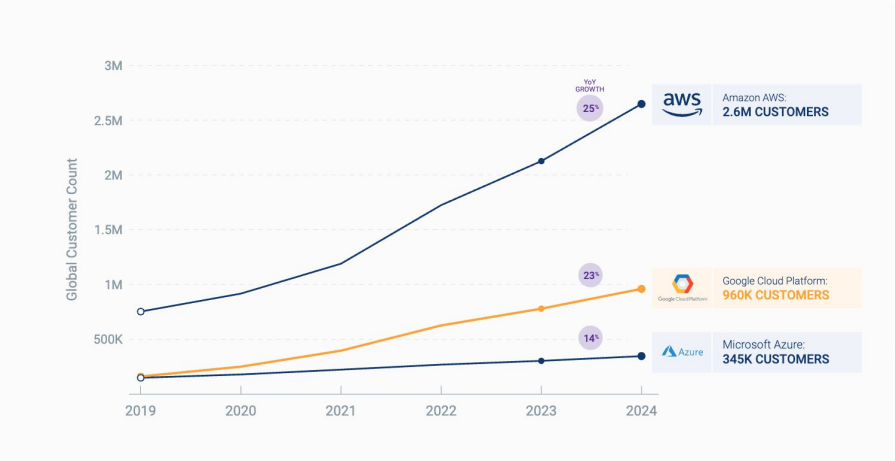
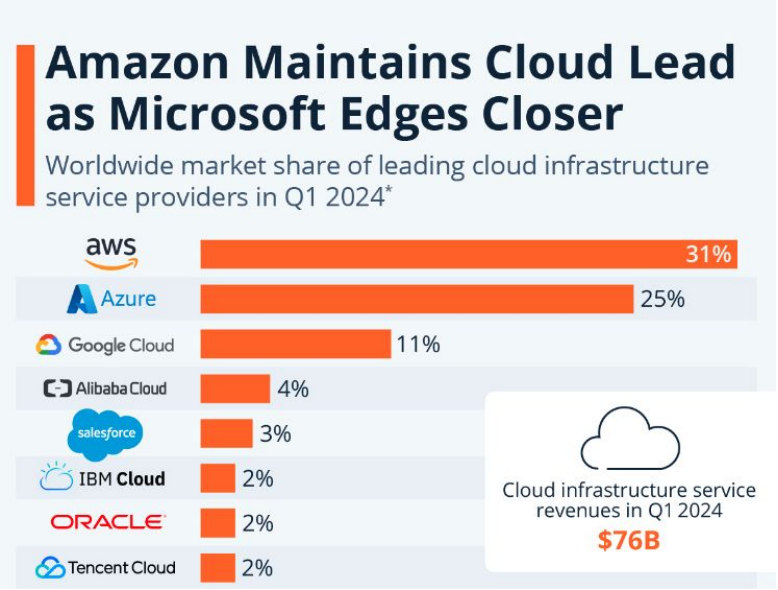
Which cloud is the best ?

Growth of Leading Providers' Customer Count Since 2019



Source: HG Insights data



Statista, 2024, Cloud Infrastructure Market

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Why GCP and not AWS ?

A high quality documentation
A nice and easy to use User Interface
Cloud Providers frameworks and languages are the same:

- Airflow
- PostgreSQL
- Kubernetes
- …

The Cloud providers are the same, different names with slightly varying features

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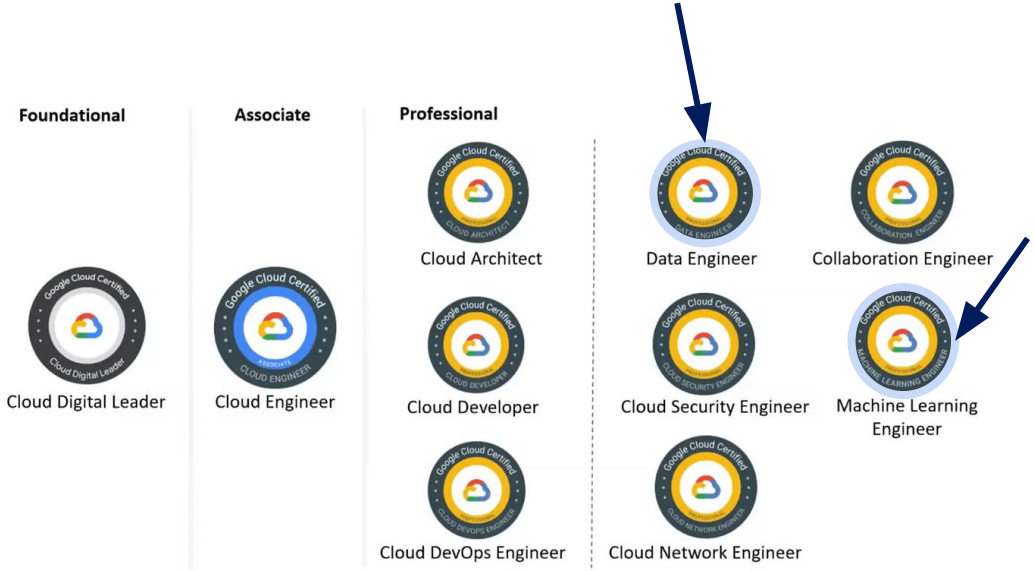
Certification

GCP offers 50% discount for certifications
Very very important to justify your skills
Nice to have in your CV

Your goal: get the
Machine Learning Engineer Certification

[About the Certification]
[Training path offer by Google]
[Question examples]



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Question

You were asked to investigate failures of a production line component based on sensor readings. After receiving the dataset, you discover that less than 1% of the readings are positive examples representing failure incidents. You have tried to train several classification models, but none of them converge. How should you resolve the class imbalance problem?

A. Use the class distribution to generate 10% positive examples.

B. Use a convolutional neural network with max pooling and softmax activation.

C. Downsample the data with upweighting to create a sample with 10% positive examples.

D. Remove negative examples until the numbers of positive and negative examples are equal.

Answer explanation

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Question

You need to build classification workflows over several structured datasets currently stored in BigQuery. Because you will be performing the classification several times, you want to complete the following steps without writing code: exploratory data analysis, feature selection, model building, training, and hyperparameter tuning and serving. What should you do?

A. Configure AutoML Tables to perform the classification task.

B. Run a BigQuery ML task to perform logistic regression for the classification.

C. Use AI Platform Notebooks to run the classification model with pandas library.

D. Use AI Platform to run the classification model job configured for hyperparameter tuning.

Answer explanation

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Half-project

Rate Chatbots with you own questions | Rate Chatbots with random questions | Chatbot Ranking

What is your question?

How many training example are required for SFT ?

Ask question

There's no single answer to how many training examples are needed for successful Supervised Fine-tuning (SFT) of a language model. The ideal number depends on several factors:

- **Model size and architecture:** Larger, more complex models generally require more data.
- **Task complexity:** Simple tasks might need fewer examples than nuanced or subjective ones.
- **Data quality:** High-quality, representative data is more valuable than a large quantity of noisy or irrelevant data.
- **Desired performance level:** Higher accuracy and generalization often require more training examples.

Rules of thumb:

- **Start small and experiment:** Begin with a few hundred or thousand examples and gradually increase the dataset size, monitoring performance improvements.
- **Consider data augmentation:** Techniques like paraphrasing or back-translation can artificially increase your training data size.
- **Look at similar tasks:** Research papers and benchmarks for similar NLP tasks can provide insights into typical dataset sizes.

In essence, it's about finding the right balance between data quantity and quality to achieve your desired performance level within your computational constraints.

There's no single answer to how many training examples are required for successful Supervised Fine-Tuning (SFT) of a language model. The ideal number depends on several factors:

Factors Influencing Training Data Requirements:

- **Model Size:** Larger, more complex models generally benefit from more training data.
- **Task Complexity:** Simple tasks (e.g., sentiment analysis) might require fewer examples than complex tasks (e.g., nuanced dialogue generation).
- **Data Quality:** High-quality, representative data is more valuable than a large quantity of noisy or biased data.
- **Desired Performance:** Higher accuracy and generalization often require more training examples.

General Guidelines:

- **Start Small, Scale Up:** Begin with a smaller dataset (hundreds to a few thousand examples) and experiment with increasing the size.
- **Consider Data Augmentation:** Techniques like paraphrasing or back-translation can artificially increase your training data size.
- **Evaluate Regularly:** Monitor performance metrics on a held-out validation set to find the sweet spot between data size and model performance.

In Practice:

- You might see good results with as few as a few hundred examples for simpler tasks and smaller models.
- Complex tasks and larger models often require tens of thousands or even millions of examples.

Key Takeaway: Focus on high-quality data and experiment to find the optimal training data size for your specific SFT task and model.

Which model do you prefer?

Left | Equal | Right

Validate preference

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Half-project

Rate Chatbots with you own questions | Rate Chatbots with random questions | Chatbot Ranking

Model Name	Students questions ELO	Template questions ELO	LLM ELO
benchmark	1011	1014	1000
florian bastin	1001	1001	1000

Refresh Ranking

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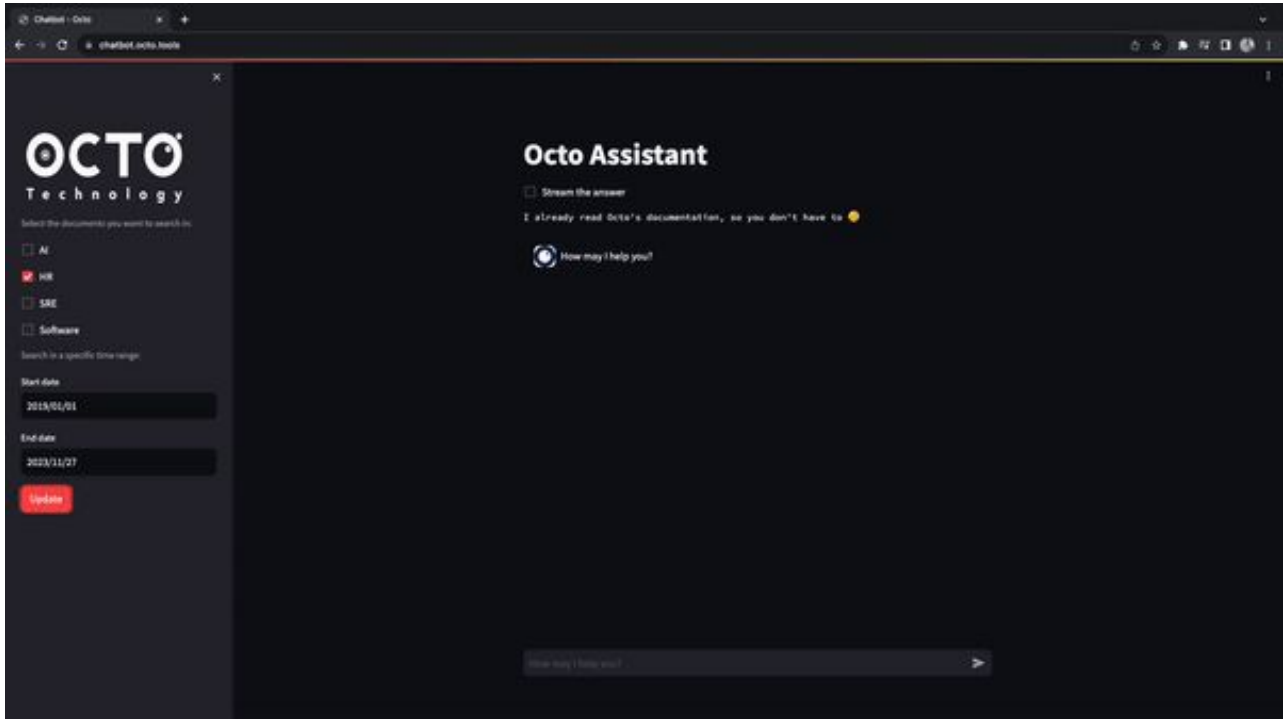
Chatbot example



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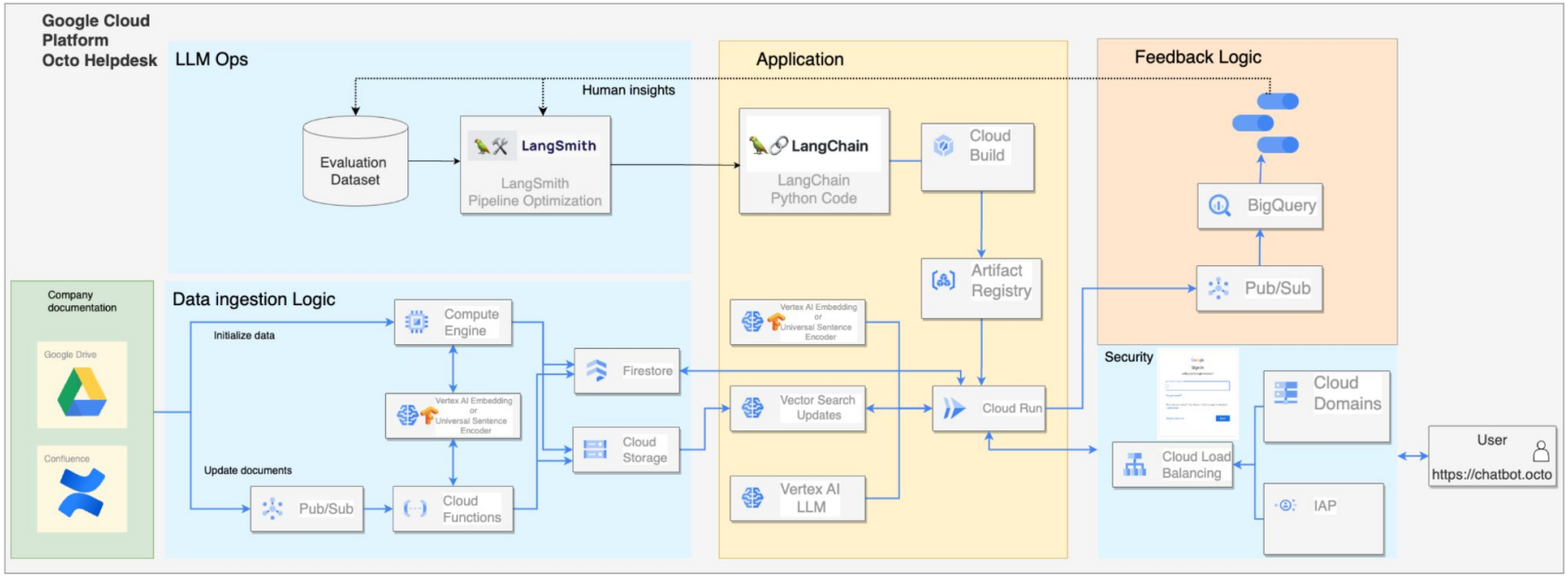
A RAG architecture in GCP



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The project

- How to create a RAG / Agent architecture ?
- How and where to store our Data ?
- How to create a frontend application in Python (Streamlit, Gradio, Chainlit) ?
- How to create a Docker image containing the Python code ?
- How to deploy an application to access the frontend using internet ?
- How to secure the application ?
- How to add multimodal data processing ?
- How to get feedback from the user ?
- Where to store the feedback ?
- How to create a feedback dashboard ?
- How to evaluate our chatbot ?

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Prerequisites

- A computer
- VS Code
- Gcloud Command Line Interface (CLI) installed
- Cloud SQL Proxy (CLI) installed
- You have received an accepted to join my GCP organization
- You have a Git Account
- You have cloned the repository

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